

Global Facilities - Design & Construction Standard - BIM - BIM Requirements for Design / Construction / As-Built

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1.0 Purpose

As BIM is required to design, construct, and maintain Micron facilities, this document is to specify the BIM requirements and guidelines for external partners not limited to, architectural and engineering (AE) firms, construction managers (CM), general contractors (GC), all other sub-contractors, and vendors who are required the use of BIM in performing facilities related work on Micron sites.

2.0 Scope

This document is applicable to all facilities related work on Micron sites requiring BIM.

3.0 Definitions and Acronyms

Term/Acronym	Definition / Description
3D	Three Dimension
ACC	Autodesk Construction Cloud
AE	Architectural and Engineering
BEP	BIM Execution Plan
BIM	Building Information Modeling

Term/Acronym	Definition / Description
BOD	Basis Of Design
CAD	Computer-Aided Design
CDE	Common Data Environment
CM	Construction Management/Managers
GC	General Contractor
IFC IFC	Issue For Construction (Software format) Industry Foundation Classes
LOD	Level of Detail (Level of Design/ Development)
O&M	Operations & Maintenance
OSM	Off Site Manufacturing
P&ID	Piping & Instrumentation Diagram
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKM's
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team

4.0 Roles and Responsibilities

Role	Responsibilities
GFTT Lead (Discipline Engineer)	• Develop and maintain guidelines. • Perform periodic audits to ensure standards set are maintained at execution. • Document and incorporate all lessons learned into future BEP templates and standards.
FCT Lead (Discipline Engineer)	• Ensure Site Construction team draft out the contents as required in the BEP template prior to project start. • Work with Site Construction team to work out areas that require customization and feedback to GFTT BIM Lead. • Perform QA/QC checks on the BIM deliverables and ensure execution as per BEP.
AE Team	• Draft contents of the Design stage BEP relevant to the project base on BEP template issued by Micron. • Execute BIM process according to BEP.
GC Team	• Draft contents of the Construction stage BEP relevant to the project base on BEP template issued by Micron. • Execute BIM process according to BEP.
Site Facilities Teams	• Verify at close-out the standards documented in BEP are adhered to.

5.0 Purpose of BIM

- To eliminate redesign at construction.
- To ensure constructability by eliminating clashes at the design and pre-construction stages.
- To enable modular and OSM design.
- To utilities for site safety assessment wherever possible.
- Ensure facility O&M requirements are considered early in the design.
- To construct 3D BIM models with information that is digital twin ready.
- Have 3D BIM models to be the source for accurate spatial information.

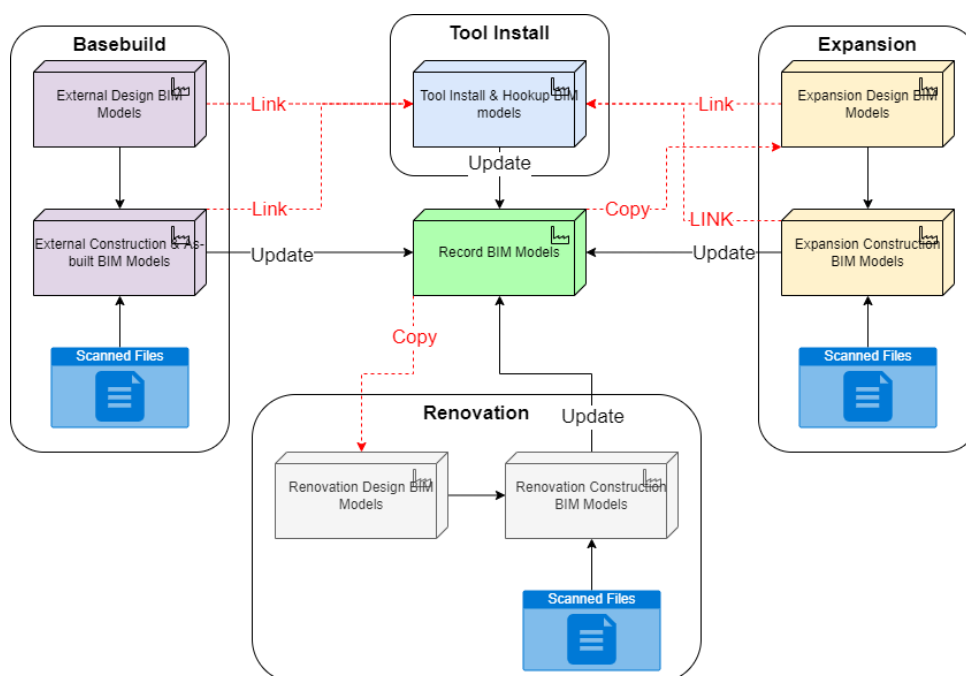
6.0 Strategy

- BIM to start at design.
- BIM requirements to be made known to partners at the start of the design.
- BIM Execution Plan (BEP) to be prepared to the level ready for execution at the early stage of the project by following the Micron BEP template provided.
- Get involvement of stakeholders at the appropriate times to provide inputs into the design with the aid of BIM models.
- Ensure the accuracy of the BIM models to the as-constructed condition as much as possible with the aid the best available technologies.
- Have all data to link to the BIM models to be ready for integration into the digital twin.
- Ensure the BIM models and information are always up to date.

7.0 BIM Models

7.1 Record Model

- Is a living Record Model for the lifecycle of the facility.
- It is to be owned, updated, and maintained by Micron BIM team in each site.
- It is the base reference model for all works to be done to the existing facilities.



7.2 Design Model

- Planning, design, engineering, and coordination.
- Model setup may vary slightly depending on project approach; however, all model participants performing design-related activities must adhere to all requirements defined in this BIM Specification and the AE's BEP.
- 3D laser scanning will be required for projects that are to be executed within an existing facility as part of the design process. Refer to the 3D Laser Scanning requirements section in this document.
- Models are to be used for quantity take-off and estimating.
- Work in progress models can be provided for early tool layout and hookup planning where needed.
- Design coordinated models are to be used for OSM and modularization concept planning.

7.3 Construction and As-Built Model

- Pre-construction planning, construction, installation, and coordination.
- Aid in field and shop drawings fabrication and procurement.
- Model setup may vary slightly depending on project approach; however, all model participants performing construction-related activities must adhere to all requirements defined in this BIM Spec and the GC or CM's BEP.
- 3D laser scanning is required and is an integral part of the construction process. Refer to the 3D laser scanning requirements section in this document.
- Models are to be updated to as-constructed/as-built condition using laser scans to provide for early tool layout and hookup planning.
- Models are to be used for quantity take-off and cost validating and verification.
- As-built models with data collected for integration to future facility O&M needs.

8.0 BIM Modelling Standards

8.1 External Model Participant Requirements

- AE, GC/CM, and all model participants must read, comprehend, and adhere to this BIM Specification and associated attachments.
- All requests for deviation from this BIM Specification and associated attachments must be pre-approved by Micron BIM Leads assigned to the project and clearly defined in the AE and GC/CM's BIM Execution Plan
- AE and GC/CM (as applicable to the project) are required to submit a BIM execution plan (BEP) for review and approval by Micron BIM Leads assigned to the project prior to commencement of work.
- The BEP must demonstrate adherence to this BIM specification and associated attachments (and any approved deviations).

8.2 Model Software Requirements

- Below are the preferred software and collaboration platform/s to use. AE or GC/CM may provide recommendations and provide to Micron for assessment and approval.

USE	SOFTWARE / PLATFORM	FILE FORMAT	VERSION
Design, Construction Authoring	Autodesk Revit	rvt	2020*
Model coordination	Autodesk Navisworks Simulate or higher BIM360 / Autodesk Construction Cloud (ACC)	nwc, nwd, nwf All other formats that can be view in the platform	2020*
Clash Detection, Monitoring, Tracking & Reporting & Review	Autodesk Navisworks Manage BIM360 / Autodesk Construction Cloud (ACC)	nwc, nwd, nwf All other formats that can be view in the platform	2020*
2D Drafting	Autodesk Autocad	dwg	2010 or higher

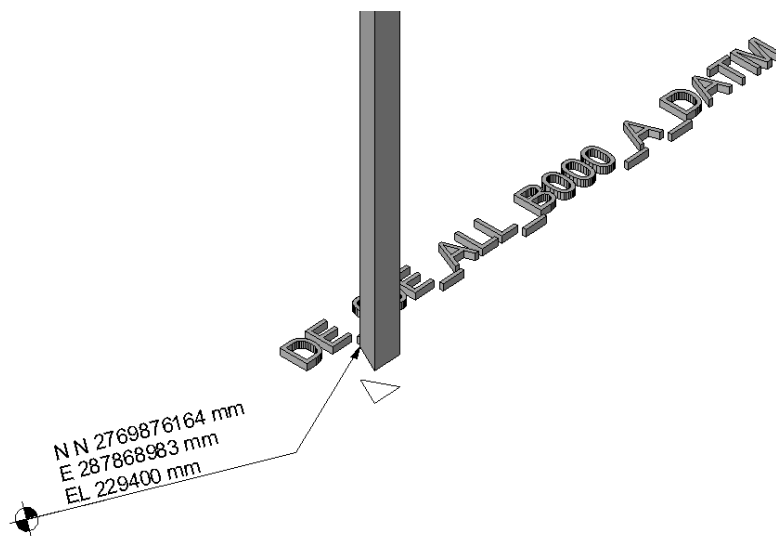
* The version of the softwares used will be decided by micron during the project kickoff and will depend mostly on the time of execution of the project.

8.3 Model File Naming

- Model naming standard is to adhere to Micron standards as indicate in BIM Model and File Naming Convention attachment.

8.4 Model Positioning and Alignment

- A common model containing all the datum information; site survey points, project base points and levels of buildings are to be created by the AE and GC/CM.
- The model is to be used to link all building, structures, site topography, etc. for alignment.
- The model can also be used for metadata checking and extraction.
- A common alignment point is to be established and marked with a triangular column as shown below that is to be placed in all the models using the same coordinates. This is to be used as a quick check and alignment when federating the models. The name of this column is to be the same as the model that is it placed in.



8.5 Model Standards

8.5.1. Title block

- Refer to section 15.

8.6 Model Quality Control

- AE and/or GC/CM are to perform model quality check and maintained from model creation to model handover.

- The following are recommended quality checks. Additional checks may be added in the BEP where required.

Checks	Definition
Visual	Ensure model components adhere to LOD and that BOD and/or Design is being followed
Interference	Monitor all model participants for adherence to clash coordination expectations via Navisworks
Standards	Ensure that BIM and CAD standards have been followed (fonts, dimensions, layers, work sets, etc.)
Model Integrity	Compare all models against BOD/Design, P&IDs, and Construction Specs. Ensure all content adheres to listed documents and/or approved submittals
Appended Model Workflow	All model participants are required to append their Revit models into Navisworks real-time, to ensure coordination with all other model participants.

- Below are the mandatory checks to be carried out on a frequently from the start:
 1. Model Naming
 2. Model Position and alignment

8.7 Reality Capture and Model Update

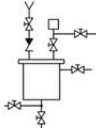
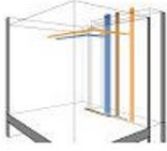
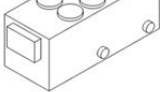
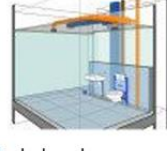
- Reality capture is required to ensure accuracy of new construction and existing conditions. Use of 3D scans, or any similar technology available at the time, will vary depending on project type and approach (i.e., new greenfield construction, remodel/retrofit).
- A scanning plan with timing, frequency and priorities shall be clearly defined in the AE and/or GC/CM BEP (as applicable to the project) and approved by the Micron BIM Leads assigned to the project.
- Scanned conditions are to be compared with the BIM models and appropriate actions to be taken (ie, adjust the BIM models or site adjustments to follow the coordinated design performed in the BIM models).
- Final scan of the as-built conditions must be done and updated into the BIM models.

9.0 BIM Model Contents

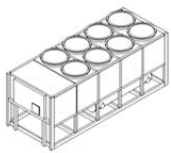
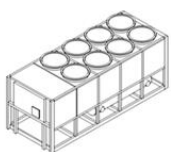
9.1 Model Level Of Development (LOD) Requirements

- The LOD defined here is a combination of the level of geometric detail and level of information to create at each stage for its intended purpose.
- The LOD required is progressive throughout the design, construction and reaches a level of 500, where information, or metadata, entered must be sufficient for the purpose of facility management.
- The following are the minimum requirements and act as a guide for LOD development. Additional requirements may be added in the BEP where needed.

9.1.1 Design Stage

Stage	CSA	MEP	Equipment	Level of Information (LOI)
Concept Design (30% Design) BIM100	 Include columns, floors, walls and openings (approximate size).		 	Graphical and function information.
Schematic Design (60% Design) BIM200	 Include beams and openings sizes to design requirements	 Vertical main, gravity pipes and equipment move-in paths.	 	Approximate dimension and design information (e.g., flow rates, capacity, etc.).
Detail Design (90% & 100% Design) BIM300		 All MEP distribution services.	  Equipment selected at design.	More accurate dimension, design information and estimated quantities.
Construction Documentation (IFC) BIM350		 Include only common & critical supports.	  Equipment based on defined specifications	Dimension and design information according to specifications.

9.1.2 Construction to As-Built Stage

Stage	CSA	MEP	Equipment	Level of Information (LOI)
Preconstruction BIM350		 Includes all supports & items for construction.	 Actual equipment procured.	Coordination information, work sequence, equipment information(e.g., manufacturer, model type, etc.) and preconstruction estimate quantities.
Construction BIM400		 Fabrication/Shop drawings.	 Actual equipment to be installed.	Work sequence, actual constructed/installed information(e.g., Location, manufacturer, equipment name, serial numbers, etc.). Information need to enter throughout construction. Quantity tracking.
As-built BIM500		 Verified with actual site condition.	 Actual equipment installed.	Full accurate information(e.g., Location, manufacturer, equipment name, serial numbers, etc.), PLUS operations required information(e.g., operation manuals, warranties, maintenance requirements, etc.). Quantity verification.

- All systems in project scope are to be modeled regardless of size. Exceptions must be reviewed and approved by the Micron BIM Leads assigned to the project and called out in the AE and/or GC/CM BIM Execution Plan.
- All models will be updated and maintained throughout construction to reflect all approved field updates, laser scans, RFI's, change orders, and design updates/revisions. The AE and/or GC/CM will assign model update responsibility in their BIM Execution Plan.
- All 3D content added to the design model and the construction model (by all model participants and all disciplines) will adhere to the LOD as indicated.
- Internal workings and parts of equipment, not affecting construction, installation or coordination with other services, may be modelled at a low level of LOD or not be modelled unless agreed with Micron and stated in the GC/CM BIM Execution Plan.

9.2 Model Metadata Requirements

- Metadata refers to non-graphical information about an object which help to define, not limited to, its dimension, location, material, technical data for design decisions, technical data of the actual installed/constructed for future facility O&M needs.
- All required metadata must be included with model content as it is available and added to the model and the actual object installed/constructed to update into the model.
- The following are recommended metadata to add at each stage of the design and construction. Additional may be added in the BEP where required.

9.2.1 Design Stage

Stage	CSA/MEP	MEP	Remarks
Concept Design (30% Design) BIM100	• Space information		
Schematic Design (60% Design) BIM200	• Space information • Design data	• Design data	
Detail Design (90% & 100% Design) BIM300	• Space information • Design data • Dimensions • Material information	• Design data • Dimensions • Material information	
Construction Documentation (IFC) BIM350	• Space information • Design data • Dimensions • Material information	• Design data • Dimensions • Material information • Location (Grids, level, room, etc.)	Final design updates.

9.2.2 Construction to As-Built Stage

Stage	CSA/MEP	MEP	Remarks
Preconstruction BIM350	Update as available	Update as available	
Construction BIM400	Update as available	Update as available	
As-built BIM500	• Space information • Design data updated to actual/constructed	• Location (Grids, level, room, etc.) • Design data updated to actual/constructed	

Stage	CSA/MEP	MEP	Remarks
	- As-installed product data - Actual dimensions - Material information - Item serial number - Manufacturer	- As-installed product data - Actual dimensions - Material information - Item serial number - Manufacturer - Technical data, etc	

9.3 Model Components/Families

- When creating Revit components, correct family categories must be used as much as possible. However, in the case where appropriate categories are not available, other categories may be used provided that the metadata can be shown within the category used.
- When using replacement categories, the intended use category and discipline are to be indicated in the component name. (Refer to BIM Model and File Naming Convention attachment)
- All third-party geometry must be Revit native content. Families derived from non- Revit platforms must be fully converted to Revit .rfa files, and clear of any CAD import contents.
- All equipment maintenance clearances are to be created.
- Model-In-Place families shall not be used.
- Conduit category shall not be used. Pipe category shall be used in place of it.

9.4 Reality Capture and Model Update

- Reality capture is required to ensure accuracy of new construction and existing conditions. Use of 3D scans, or any similar technology available at the time, will vary depending on project type and approach (i.e., new greenfield construction, remodel/retrofit).
- A scanning plan with timing, frequency and priorities shall be clearly defined in the AE and/or GC/CM BEP (as applicable to the project) and approved by the Micron BIM Leads assigned to the project.
- Scanned conditions are to be compared with the BIM models and appropriate actions to be taken (ie, adjust the BIM models or site adjustments to follow the coordinated design performed in the BIM models).
- Final scan of the as-built conditions must be done and updated into the BIM models.

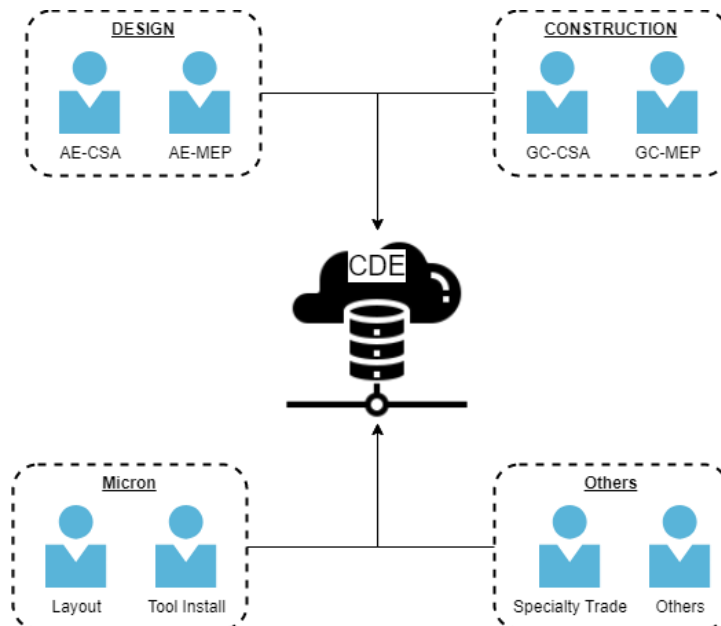
10.0 Model Collaboration

10.1 Overview

- Model collaboration is an integral part of the BIM process.
- The following will provide guideline on model sharing, model storage and clash management.

10.2 Model Sharing and Storage

- AE or GC/CM is required to provide or use a Common Data Environment (CDE) for stored and exchange of BIM documents as specified by Micron as indicated in section 4.2 Software requirements. AE or GC/CM may provide recommendations on the CDE to be used and provide to Micron for assessment and approval.

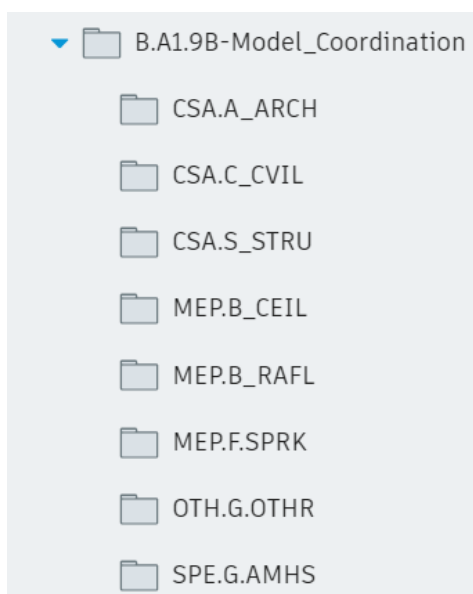


- All model participants, not limited to AE, GC/CM, trader contractors, equipment vendors are to store and exchange all BIM documents, not limited to models, drawings etc., using the CDE.
- AE and/or GC/CM are to consolidate, federate and upload the models by all model participant into a commonly viewable and exchangeable file format as indicated in section 4.2 Software requirements regular basis as agreed by all stakeholder and stated in the BEP.
- The AE and/or GC/CM must maintain a common collaboration model file/s (e.g., Navisworks nwf), where applicable, for documentation and recording of all clash analysis viewpoints, standard project views, etc.

- Common collaboration model files shall be stored in a folder clearly indicating it as a federated model file location and separated by file formats (an example as shown below).



- Each model participant shall store their work in progress BIM models in folders created in the CDE indicating if they belong to CSA, MEP, Specialty trades or others, main discipline, sub-discipline, and trades. Access permission be control by Micron (an example as shown below).



- Folder structure and BIM document exchange workflow shall be included in the BEP.

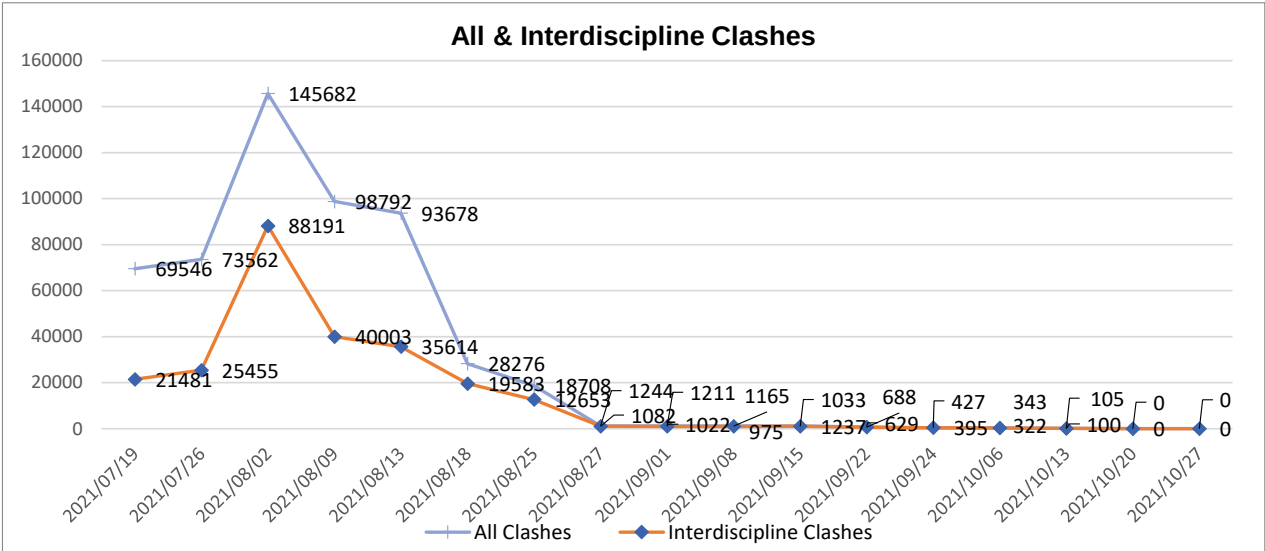
10.3 Clash Management

- AE or GC/CM BIM leads, or BIM Coordinators are to perform the tasks of clash detection, coordination and the clash resolved in a timely manner.
- Clashes shall be communicated to all the model participants, with the best know method and/or available technology, in a real-time manner.
- AE or GC/CM BIM leads, or BIM Coordinators are to hold regular meetings with all model participants and stakeholders to review, discuss and resolve the clashes to meet the project deadlines.
- Issues, viewpoints, created, not limited to clashes, are to be recorded and tracked using the best-known tools/technology available. Where possible, this information is to be tracked and recorded within the CDE platform if available.
- A clash rule table is be provided to clearly indicate how the clashes are to be setup in Navisworks or any clash detection software or platform to be used for the purpose.

Rule Names in Navisworks	Clash Item 1	Clash Item 2	Clash Type
A01_E vs E	All models with "_E_" discipline code	All models with "_E_" discipline code	Same discipline
A02_E vs M	All models with "_E_" discipline code	All models with "_M_" discipline code	Interdiscipline
A03_E vs P	All models with "_E_" discipline code	All models with "_P_" discipline code	Interdiscipline
A04_E vs F	All models with "_E_" discipline code	All models with "_F_" discipline code	Interdiscipline
A05_E vs H	All models with "_E_" discipline code	All models with "_H_" discipline code	Interdiscipline
A06_E vs T	All models with "_E_" discipline code	All models with "_T_" discipline code	Interdiscipline
A07_E vs Y	All models with "_E_" discipline code	All models with "_Y_" discipline code	Interdiscipline

- AE or GC/CM BIM leads, or BIM Coordinators are to provide regular reporting on the clash management status with clash matrixes, trends graphs, any other forms that can very clearly show the status and actions to be taken to resolve them (examples as shown below).

All	E	M	P	F	H	T	Y	D	N	I	Q	G	B	A	S	C	MEP	CSA	All_Clashes	Same_Discipline_Clashes	Interdiscipline_Clashes
E	111	0	4	162	0	0	0	135	0	100	786	0	0	1087	3217	1	0	0	65169	7148	58021
M		4031	391	5580	0	0	0	2114	0	233	15	0	0	2503	2559	167	0	0			
P			203	0	0	0	0	0	0	26	0	0	0	2652	2038	24	0	0			
F				983	0	0	0	498	0	300	36	0	0	2382	5160	0	0	0			
H					0	0	0	0	0	0	0	0	0	0	0	0	0	0			
T						0	0	0	0	0	0	0	0	0	0	0	0	0			
Y							0	0	0	0	0	0	0	0	0	0	0	0			
D								413	0	16	0	0	0	429	778	0	0	0			
N									0	0	0	0	0	0	0	0	0	0			
I										1	44	0	0	180	72	0	0	0			
Q											1406	0	0	0	0	0	0	0			
G												0	0	0	0	0	0	0			
B													8541	6675	9116	0	0	0			
A														4384	19088	148	0	0			
S															20432	434	0	0			
C																250	0	0			
MEP																	0	0			
CSA																		0			



- Details on clash management strategies, methodologies, workflows, etc. shall be included in the BEP.

11.0 BIM Deliverables

- AE, GC/CM shall provide a complete BIM Execution Plan (BEP) 30 days of the letter of award. A draft BEP shall be submitted by AE, GC/CM at BIM kickoff meeting to start off the review process. Regular meetings shall be conducted by the Contractor to ensure it meets the Micron requirements. A complete BEP shall be submitted to Micron for approval in writing prior to commencing work.
- All design and as-built models must be delivered clash adjudicated depicting all scope and standards as specified in this document and attachments.
- All construction and as-built models must be provided to Micron in the native software format no later than the project specified model turnover date(s). Models will include all installed scope for all. Models delivered are required to be validated and updated against final as-constructed / as-built conditions with 3D laser scans, or any accurate 3D capture technology available, prior to model turnover to Micron.
- All content in the models must be native Revit geometry only. Any non-native Revit third party software must be submitted to Micron BIM team prior to use. However, non-native Revit third party software is generally not permitted. In the case, where Revit generated geometry cannot be produced, IFC format together with 3D dwg and native formats are to be submitted.
- All 2D layouts must be produced from the 3D models and within the 3D authoring software. All 2D drawings that are not produced from 3D (eg, Schematic drawings, typical details, quantity schedules, etc) are to be drawn in the 3D authoring software (no exceptions).

No.	Items	Software Type	Format
1.	Updated BIM Execution Plan	PDF, Word	.doc, .pdf
2.	3D models - Revit models	Revit	.rvt
3.	3D models - Federated models	Navisworks	.nwd, .nwc
4.	3D models - All other models in native editable formats	Revit (preferred), IFC, 3D Autocad, Native format	.rvt, .ifc, .dwg(3D), native format
5.	Shared parameter file	Native format	.txt
6.	Parameter list	Excel	.xlsx
7.	2D drawings	Revit (preferred), Autocad, PDF	.rvt, dwg, .pdf, .dwfx

12.0 RFP BIM Requirement for Design

1. BIM Manager to be appointed as part of the AEs design team and will be the point of contact for BIM related matters.
2. BIM Execution Plan (BEP) shall be required to be drafted at the start of the project covering BIM execution for design stage. The Contractor shall provide a complete BIM Execution Plan (BEP) 30 days of the letter of award. A draft BEP shall be submitted by the contractor at BIM kickoff meeting to start off the review process. Regular meetings shall be conducted by the Contractor to ensure it meets the Micron requirements. A complete BEP shall be submitted to Micron for approval in writing prior to commencing work on the LOD350 Model.
3. The BEP is a live document and is expected to be maintained and updated throughout the project and to submit as part of BIM deliverables prior to handing over to the construction team.
4. AE shall prepare the BIM deliverables as indicated in the RFP and the BIM Execution Plan.
5. All design drawings are to be done in BIM using Revit as primary 3D authoring software from start of design.
6. 2D drawings must be produced directly from BIM Models and within the 3D authoring software. And issued 2D drawings and 3D models must match.
7. 3D models shall be expected to be done to the level of detail and information of LOD350 or as agreed in the BEP.
8. BIM must be considered and included in the project execution workflow/processes.
9. Coordinated BIM models are to be used for and not limited to, modular, prefabrication, off-site-manufacturing/construction (OSM/OSC) studies to provided recommendations.
10. Lead AE will be required to consolidate and federate all BIM information from all disciplines and trade vendors (eg, AMHS, etc) into the BIM model.
11. Design level space management are to be done with FM operations team input.
12. Design coordination and clash detection are to be done prior to producing "Issued For Construction (IFC)" documentations.
13. Final BIM deliverables to be delivered at the end of the design project are as referenced below or otherwise agreed in the BEP.

No.	Items	Software Type	Format
1.	Updated BIM Execution Plan	PDF, Word	.doc, .pdf
2.	3D models - Revit models	Revit	.rvt
3.	3D models - Federated models	Navisworks	.nwd, .nwc
4.	3D models - All other models in native editable formats	Revit (preferred), IFC, 3D Autocad, Native format	.rvt, .ifc, .dwg(3D), native format
5.	Shared parameter file	Native format	.txt
6.	Parameter list	Excel	.xlsx
7.	2D drawings	Revit (preferred), Autocad, PDF	.rvt, dwg, .pdf, .dwfx

14. AE is expected to reference in all disciplines to commence design development.
15. AE is expected to share models of their discipline as requested by Micron.
16. Any existing building/services that interface with the project shall be modelled unless otherwise stated by Micron.
17. Coordinated design BIM models are part of the deliverables required for design.
18. 3D BIM models shall be use for not limited to, space review with stakeholders, prefabrication, constructability studies and aid in EHS assessment.
19. Appointed lead AE shall be responsible for consolidating and managing all BIM models including, and not limited to, Component & Equipment Models by trade subcontractors, vendors, suppliers, etc., who are engaged either directly or indirectly by the AEs.
20. Successful bidder shall provide BIM360 platform and licenses, as the Common Data Environment (CDE) for and not limited to drawing, BIM model exchange, design review, markups, comments, viewing, clash management, BIM deliverable submissions, etc.
21. The CDE shall allow direct access (i.e., Revit Cloud Work-sharing) to the work in progress BIM models by all collaboration partners for real-time collaboration and successful bidder shall provide sufficient licenses to allow access.

22. Successful bidder may provide Micron any solutions available at the time of, or after the project award that will benefit design and effective collaboration with all stakeholders (e.g., AR, MR, 2D and 3D overlay for viewing in mobile devices, etc.) for use in the project.
23. All models delivered to Micron must be in Revit format and other formats as defined in the BEP.
24. Micron shall have ownership and exclusive rights to all BIM deliverables and usage for the project. And may grant access to use the BIM deliverables in successive operations and Maintenance works at Micron's discretion.
25. All OSM modules and assemblies shall have their own Design BIM models which are coordinated and interfaced with the Project and Site consolidated BIM Model, for dimensional verification, interface coordination, clash detection, etc.
 - a. Unless otherwise noted, OSM modules and assemblies shall include 3D Laser Scanning as-built verification at the OSM supplier site, before shipping and delivery to Micron Site. The Micron Site actual field conditions shall also include 3D Laser Scanning as-built verification prior to shipping and delivery of the OSM modules to Micron Site. The 3D Laser Scanning of OSM modules & assemblies, 3D Laser Scanning of Micron Site actual field conditions, and the BIM consolidated model shall be reviewed together for dimensional verification, interface coordination, clash detection, etc.

13.0 RFP BIM Requirement for Construction to As-built

- All models delivered to Micron must be in Revit format and other formats as defined in the BEP.
- GC to receive LOD350 Design BIM models from AE and require to further develop and/or modify to LOD 500 As-Built BIM models at close-out.
- In the case, no Design BIM models from AE are provided, GC will be expected to create LOD350 Construction BIM models base on received IFC design and further develop and/or modify design to LOD400 Construction BIM models to LOD500 As-Built BIM models at close-out.
 - LOD500 As-Built BIM Models shall include 3D Laser Scanning for actual verification.
- BIM models after handover from AE, or created by GC, will act as Record BIM models and ownership will belong to the GC.
- BIM must be considered and included in the project execution workflow/processes.

- BEP is to be drafted with priority sections to be completed and pre-approved before the start of BIM execution and updated at close-out.
- All models are to be coordinated and clashes resolved prior to 2D shop drawing submissions and construction/installation.
- All changes are to be verified 3D first prior to 2D shop drawing submissions and construction/installation.
- 3D models of the area of work are to be submitted together with the 2D shop drawings for shop drawing submission and review.
- Ensure the BIM models are used for constructability studies prior to development of shop drawings.
- Ensure Space management done with FM operations team input.
- Ensure coordinated models are used to help identify areas for prefabrication and off-site manufacturing and work with trade contractors, vendors, suppliers, etc. to provide workable solutions.
- Ensure coordinated models are used to help identify EHS issues with all EHS team member input.
- Work sequence are to be studied using the coordinated BIM models prior to fixing work schedules.
- All models and elements must be verified with the 3D scans, which are to be included as part of the deliverables, and accurate to the as-built conditions at close-out.
- All BIM deliverables are to be submitted to Micron at the end of project close-out as will be agreed in the BEP.
- For areas that require tool installation and hookups, as-constructed base build site conditions are to be scanned and updated into the BIM models prior to design and installation works, even if it has not completed base build as-built stage.
- Highly recommended that 3D scans to be done after completion by single trade, discipline or subdiscipline to ensure Construction BIM models are accurate to the as-constructed conditions. Alternative technologies like AR may be used to ensure as-constructed/as-installed are according to the coordinated Construction BIM models.
- 2D layouts must be produced from the 3D models and within the 3D authoring software.
- All schematic diagrams not limited to, P&IDs, Single line, Riser diagrams, etc., are to match the 3D models and items are hyperlinked to the 3D elements.

- All required data related to the constructed or installed items, whether or not provided by the client, are to be entered into BIM models, verified, and updated into BEP at close-out.
- All related documents that come with the constructed/installed items can be referenced to the elements within the BIM models.
- Material quantities, as much as possible, to come from 3D models.
- All naming conventions, model setups, modeling standards, etc., are to follow what are stated, or update if changed, in the BEP at close-out.
- Provide BIM360 platform as the Common Data Environment (CDE) for, and not limited to, drawings, BIM models exchange, design review, markups, comments, viewing, clash management, submission, etc. The CDE shall allow “live” access (eg, Revit Cloud Worksharing) to the work in progress BIM models by all collaboration partners for real-time collaboration. GC must provide sufficient licenses for the platform use.
- GC expected to provide trainings to Micron where required.

14.0 RFP BIM Scope for Construction to As-built

1. The Contractor shall provide a complete BIM Execution Plan (BEP) 30 days of the letter of award. A draft BEP shall be submitted by the contractor at BIM kickoff meeting to start off the review process. Regular meetings shall be conducted by the Contractor to ensure it meets the Micron requirements. A complete BEP shall be submitted to Micron for approval in writing prior to commencing work on the LOD350 Model.
2. The Contractor shall be responsible for developing and maintaining the BIM Models from Design BIM Models (LOD350) to Construction BIM Models (LOD400) and the final delivery of As-Built BIM Models (LOD500) to Micron. In the case where Design BIM Models are unavailable, the Contractor will be required to produce Construction BIM Models base off the IFC drawings provided to start off the BIM Execution work.
 - LOD500 As-Built BIM Models shall include 3D Laser Scanning for actual verification.
3. The Contractor will lead, coordinate, establish protocols related to creation, review, consolidate and management of BIM models including, and not limited to, Component & Equipment Models by trade subcontractors, vendors, suppliers, etc, who are engaged either directly or indirectly by the Contractor.
4. The Contractor shall use the Construction BIM Models to study the constructability of the design and provide timely feedback to the designers to adjust the design where required.

5. The Contractor shall perform space management work on the Construction BIM Models with inputs from Micron facility operations team to optimize the arrangements of all services for construction and operations & maintenance needs.
6. The Contractor shall utilize the coordinated Construction BIM Models to identify area for prefabrication, off-site manufacturing and work with trade contractors, vendors, suppliers, etc. to provide workable solutions to increase speed of construction/installation and eliminate safety risk on-site.
7. The Contractor shall work with both internal and Micron EHS team in using the coordinated Construction BIM Models to identify areas with safety concerns to ensure a complete safety plan can be created, safety measures done on the constructions site prior to construction/installation with the aim to achieve zero incident on site.
8. The Contractor shall utilize the coordinated Construction BIM models for work sequence planning by entering construction work sequence data in the BIM models to help improve the project construction schedule.
9. The Contractor shall produce and submit shop drawings directly off the coordinated Construction BIM Models and within the 3D authoring software.
10. The Contractor will be responsible to create As-Built BIM Models which shall document actual as-built and installed conditions. The As-Built BIM Models shall also contain additional data such as equipment property data for facility management purpose and create hyperlinks to related drawings and documents.
11. The Contractor shall perform 3D scans on the as-constructed condition once complete and update the Construction BIM Models accordingly. Contractor may employ other technologies, such as Augmented Reality, to ensure the as-constructed condition is the same as the Construction BIM Models.
12. The Contractor shall enter relevant As-built BIM Models which shall include, and not limited to, descriptions, dimensions and, as applicable, manufacturer, model number, serial number, purchase & install dates, specifications, operations & service manuals, etc, and should be documented in the BEP.
13. The Contractor shall provide dedicated qualified BIM Manager and BIM staffs, to be stationed on-site where conditions permit, to develop, execute and manage all the BIM processes and BIM deliverables. Their roles and responsibilities shall be clearly defined in the BEP. The GC to provide a minimum of 1 BIM manager, 2 BIM coordinators, 1 space management with BIM capability personnel, 1 BIM quantity takeoff(5D and schedule updater(4D), 2 laser scanners, 1 BIM modellers for each discipline.

14. The Contractor shall provide and maintain a BIM Collaboration Platform as a Common Data Environment to host all BIM models and files for data exchange. BIM models and files in this platform shall be fully accessible online and allow “live” access (eg, Revit Cloud Worksharing) to the work in progress BIM models and files for real-time collaboration to all required authorized project team members to ensure speedy delivery of a Construction BIM models that will be ready to produce 2D shop drawings for construction.
15. The Contractor shall facilitate, define, and document a file structure for BIM data and file exchange in the BEP.
16. The Contractor shall provide trainings where required.
17. The Contractor shall provide BIM progress updates regularly to Micron throughout the construction phase on the coordination progress, clash detection and BIM modelling for shop drawing statuses.
18. The Contractor shall be responsible to initiate and lead meetings to achieve timely communication of information between stakeholders.
19. File size of the BIM Models shall be managed so as not to affect the work to be perform on them by the project stakeholder. The files may be split to manageable size where required.
20. All OSM modules and assemblies shall have their own Design BIM models which are coordinated and interfaced with the Project and Site consolidated BIM Model, for dimensional verification, interface coordination, clash detection, etc.
21. Unless otherwise noted, OSM modules and assemblies shall include 3D Laser Scanning as-built verification at the OSM supplier site, before shipping and delivery to Micron Site. The Micron Site actual field conditions shall also include 3D Laser verification prior to shipping and delivery of the OSM modules to Micron Site. The 3D Laser Scanning of OSM modules & assemblies, 3D Laser Scanning of Micron Site actual field conditions, and the BIM consolidated model shall be reviewed together for dimensional verification, interface coordination, clash detection, etc. Scanning as-built.
22. The Contractor shall ensure:
 - Establish, execute, and manage BIM processes that leads to creating coordinated Construction BIM Models with sufficient data to ensure construction can be carried out smoothly and aim for zero reworks.
 - Establish, execute, and manage BIM processes that leads to creating accurate as-constructed and as-built BIM models and all other deliverables for Micron to be able to use for facility operations & maintenance, and expansion purposes.

- All partner, not limited to, trades, vendors, suppliers, etc, BIM models either directly or indirectly engaged by the Contractor are consolidated and used in the BIM coordination effort.
 - Lead and conduct, not limited to, BIM Kick-off meeting, regular progress update meetings, daily coordination & clash detection sessions, workshops, trainings to all stakeholders, BEP updates and any other meetings whenever deemed necessary.
 - All constructability issues are discussed and resolved using the Construction BIM prior to construction and installation and provide actionable feedbacks and directions with the relevant stakeholders.
 - Optimize the space arrangement for, not limited to, construction, installation, future maintenance and EHS needs by initiating and collaborating with relevant stakeholders.
 - Technical proficiency of subcontractors is adequate to participate in BIM execution.
 - Participate and use the latest available Construction BIM models in Owner, Designers and Contractor team review meetings to show that BIM is fully utilized to aid construction work.
 - All modifications, Micron approved changes, as-constructed, as-built information are to be updated to the Construction BIM Model and submit to Micron monthly for audit.
23. The Contractor shall ensure that the Construction BIM Models and BIM deliverables satisfy the Micron requirements and forwarded by the contractor to Micron for verification and approval in writing prior to submitting interim payment for work completed.
- The requirements shall include and not limited to:
 - 2D drawings match 3D models and produce directly within 3D authoring software.
 - 2D schematic elements are hyperlinked to 3D elements.
 - Data entered to the 3D models.
 - 3D models must be coordinated, clash detection performed, and clashes resolved prior to construction/installation work.
 - 3D models verified with as-constructed site conditions.

Final BIM deliverables to be delivered at the end of the project are as reference below or otherwise agreed in the BEP.

No.	Items	Software Type	Format
1.	Updated BIM Execution Plan	PDF, Word	.docx, .pdf
2.	3D models - Revit models	Revit	.rvt
3.	3D models - Federated models	Navisworks	.nwd, .nwc
4.	3D models - All other models in native editable formats	Revit (preferred), IFC, 3D Autocad, Native format	.rvt, .ifc, .dwg(3D), native format
5.	Shared parameter file	Native format	.txt
6.	Parameter list	Excel	.xlsx
7.	2D drawings	Revit (preferred), Autocad, PDF	.rvt, dwg, .pdf, .dwfx
8.	Laser scan data	Native format	Native format

15.0 Reference

No.	Items	Format
1.	Global Facilities - Engineering - BIM - BEP Template	.docx
2.	Global Facilities - Engineering - BIM - Model and File Naming Convention	.pdf
3.	Global Facilities - Engineering - BIM - Revit Title Block	.rva

16.0 Document Control

16.1 Approval: GLOBAL_FAC_GFTT_APPR

16.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

16.3 Final Viewer Access

All Micron Facilities Team Members at all Sites

16.4 Retention

Adherence to the requirements specified in the
[Global Facilities - Document Control Procedures](#)

16.5 Review

Biennially (two years)

17.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	05/04/23